

Sustainability Org: A GenAI-Powered Signal Detection and Recommendation Engine

Project Work Final Presentation

Vansh Verma

Reg. No. 220911622

Information Technology

Under the guidance of

Internal Guide

Dr. Sandhya Dubey

Assistant Professor

School of Computer Engineering, MIT
Manipal

External Guide

Ms. Nancy Miglani

Coordinator (HR)

Bain Capability Network, Gurgaon

July 2026

Note: Implementation details, data-source identities, and client identifiers have been generalized or anonymized in the interest of organizational confidentiality. Figures shown are representative and illustrative.



Presentation Outline

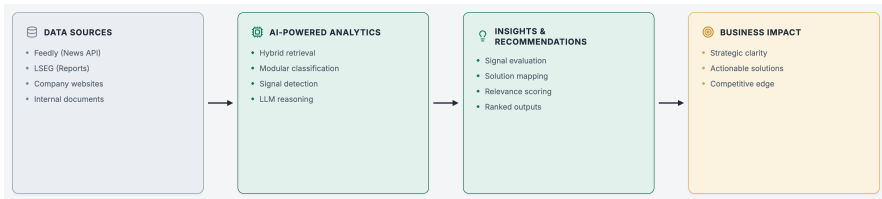
- ➊ Introduction
- ➋ Literature Survey / Background Theory
- ➌ Problem Definition
- ➍ Objectives
- ➎ Methodology
- ➏ Result Analysis
- ➐ Conclusion & Future Scope
- ➑ References



- GenAI and Large Language Models have transformed how organizations extract meaning from unstructured text.
- **Sustainability Org** is a GenAI-powered decision-intelligence engine built during an industry internship.
- It treats the problem as **signal detection**: finding latent business signals in fragmented evidence.
- Signals are **validated by retrieval-grounded LLM reasoning** and mapped to ranked, explainable recommendations.
- Designed to be **modular, explainable, and human-supervised**.



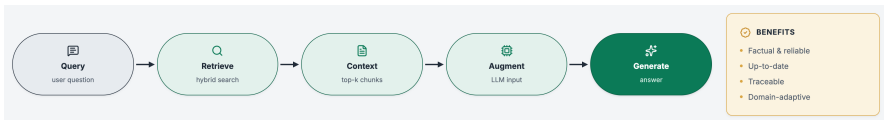
Introduction - System at a Glance



From trusted data sources, through AI-powered analytics and signal evaluation, to ranked recommendations that drive better-grounded decisions.



- **In-context learning:** LLMs can be steered for classification via prompts alone - enabling modular, YAML-configurable classification.
- **Retrieval-Augmented Generation (RAG):** grounds LLM reasoning in retrieved, citable evidence instead of parametric memory.
- **Hybrid retrieval:** fuses lexical (BM25) + dense (vector) search, then re-ranks with a cross-encoder for precision and semantic recall.



Problem Definition

- Organizations must interpret vast, fragmented, heterogeneous evidence to make decisions.
- Keyword search and rigid pipelines miss context and are hard to audit.
- Off-the-shelf LLM tools hallucinate and provide no provenance or oversight.

Goal

Transform heterogeneous, unstructured text into structured, actionable, and **explainable** intelligence - at scale, while keeping a human analyst in the loop.

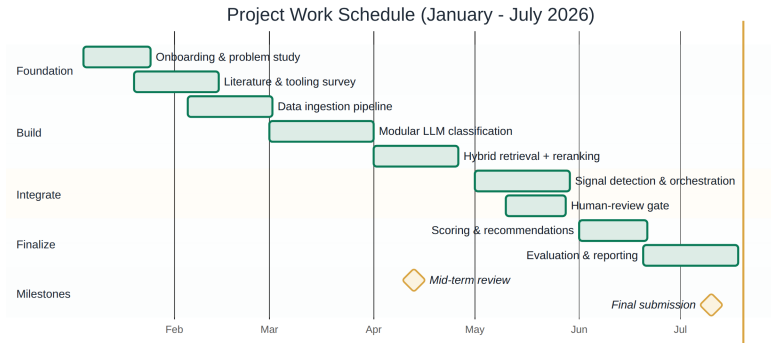


Objectives

- Build a scalable ingestion & normalization pipeline for heterogeneous sources.
- Design a modular, YAML-configurable LLM classification framework.
- Implement hybrid retrieval (BM25 + vector + cross-encoder reranking).
- Build a reasoning-driven signal-detection framework with agentic web search.
- Enforce a human-in-the-loop review gate before expensive processing.
- Map validated signals to solutions and produce weighted, ranked recommendations.



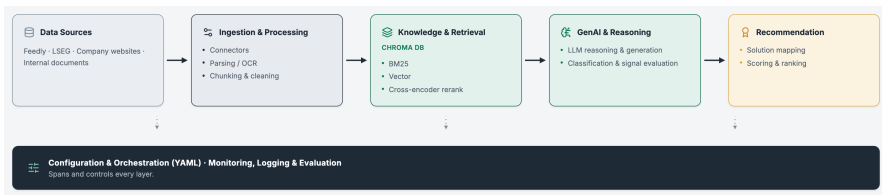
Project Timeline



Six-month internship (Jan–Jul 2026): overlapping phases from onboarding and ingestion through classification, retrieval, orchestration, and evaluation - with the mid-term review and final submission milestones.



Methodology - System Architecture

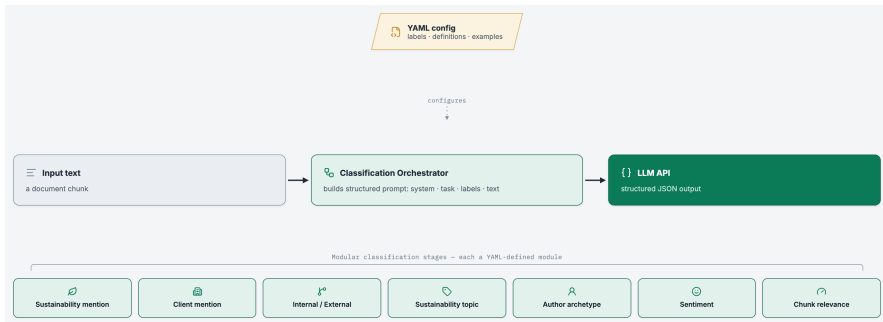


Five layers - ingestion, classification, hybrid retrieval, reasoning-based signal detection, and recommendation - all controlled end-to-end by YAML configuration.



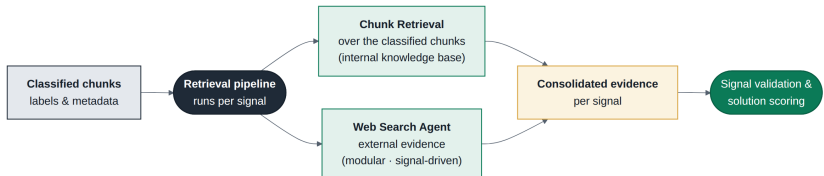
Methodology - Modular LLM Classification

- Each dimension is a configurable, prompt-driven module; executed in parallel over partitions, each label carrying a confidence score.



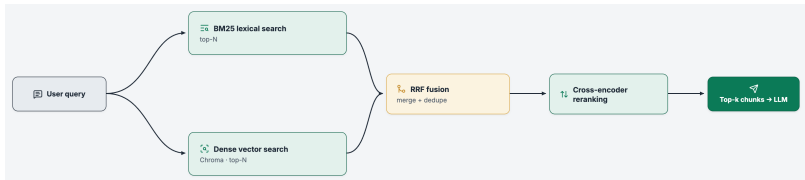
Methodology - Retrieval Pipeline (Gathering Evidence)

- Once classification is complete, the **retrieval pipeline** runs **per signal** to gather the evidence needed to validate it.
- Two complementary engines run for each signal: **chunk retrieval** over the internal knowledge base, and a **web search agent** for external evidence.
- Their outputs are consolidated per signal before validation and scoring.



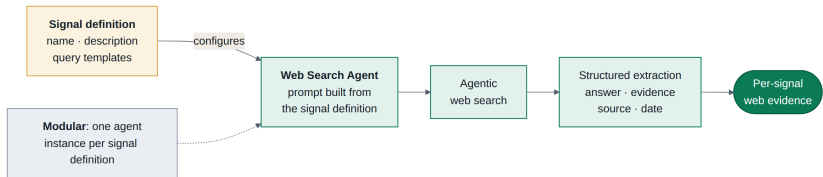
Methodology - Chunk Retrieval (over Classified Chunks)

- The **classified chunks** become the searchable knowledge base - chunk retrieval runs directly over the classification output.
- Retrieval is **scoped per signal** using metadata filters derived from the classification labels.
- Each query uses **hybrid search** - BM25 (lexical) + dense vector, fused via RRF - then **cross-encoder reranking** to return the top-*k* evidence chunks.

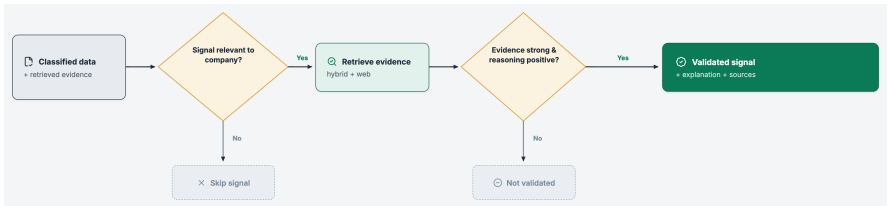


Methodology - Web Search Agent (Modular, Signal-Driven)

- Complements the internal knowledge base with **external web evidence** for each signal.
- **Modular by design**: one agent instance per signal, **configured from the signal's definition** - its name, description, and query templates.
- Runs an **agentic web search** and returns **structured evidence** - answer, supporting quote, source, and date - for each signal.



Methodology - Signal Detection & Human Review



Pre-check → evidence retrieval → LLM evaluation → validation → output. A mandatory human_check review gate excludes weak candidates before costly retrieval.



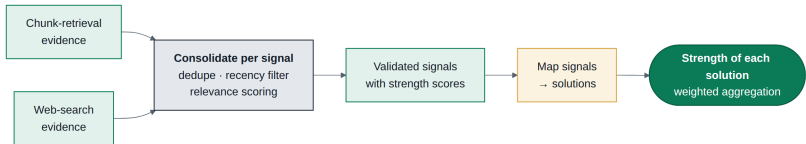
- **Scoped:** a catalogue of signals and an active run scope determine which signals participate in a run.
- **Strictly sequential:** pre-check → review → knowledge-base ingest → evidence retrieval → re-ranking - no blind parallelization.
- **Human-in-the-loop:** the run **pauses** at a mandatory review gate before any expensive retrieval is executed.
- **Resilient:** transient failures are retried; every stage is logged for auditability.

Pre-check and human review remove weak candidates *before* costly retrieval - keeping the analyst in control of every downstream conclusion.



Methodology - Consolidation & Solution Strength

- Evidence from **chunk retrieval** and the **web search agent** is **consolidated per signal** - deduplicated, recency-filtered, and relevance-scored.
- Each signal is validated and assigned a **strength score** from its consolidated evidence.
- Signals are mapped to solutions, and a **weighted aggregation** computes the **strength (fit)** of each solution.



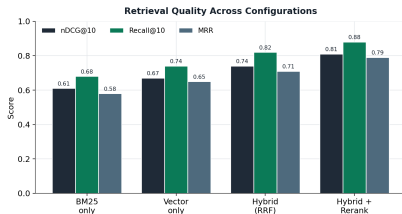
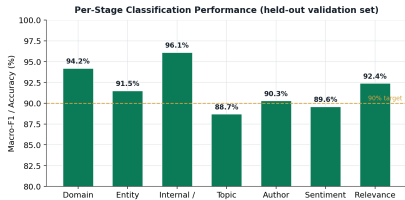
Methodology - Recommendation Engine



Validated signals are mapped to solutions and scored by a weighted engine, producing ranked, source-traceable recommendations.



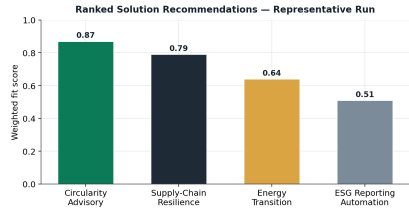
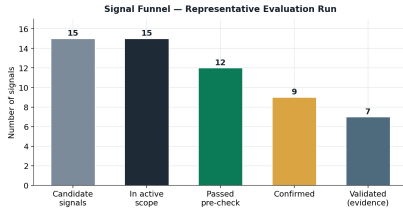
Result Analysis - Classification & Retrieval Quality



Classification mean macro-F1 $\approx$ **91%** (left). Hybrid + reranking lifts nDCG@10 from **0.61** $\rightarrow$ **0.81**, a $\approx$ 33% gain (right).



Result Analysis - Signal Funnel & Recommendations



Representative run: candidate signals narrow to a **validated subset** (left). The best-fitting recommendation scores ~ 0.87 (right) - every score traces to validated signals and supporting evidence.



Screenshots - Classification Pipeline (Notebook)

classification_pipeline.ipynb

• Python 3 — idle

+ Code ▷ Run ■ Stop ↺ Restart

```
[14]: # Run modular LLM classification across parallel partitions
      from pipeline import classification as clf
      ctx = clf.load_config("config/pipeline.yaml")
      df = clf.run_classification(raw_df, ctx, n_workers=8)
```

```
[14]: [config] company_in_focus = 'Client A'
      [engine] parallel workers = 8 ... ready
      [stage] domain_relevance ██████████ 4120/4120 38.2s
      [stage] entity_mention ██████████ 4120/4120 41.0s
      [stage] topic ██████████ 4120/4120 46.9s
      [stage] sentiment ██████████ 4120/4120 27.4s
      ✓ classification complete - 4120 rows · mean confidence 0.883
```

```
[15]:
```

	doc_id	relevance	topic	sentiment	confidence
0	DOC_0001	Yes	Energy Transition	Positive	0.94
1	DOC_0002	Yes	Circularity	Neutral	0.87
2	DOC_0003	Yes	Emissions	Negative	0.79
3	DOC_0004	Yes	Supply Chain	Positive	0.96
4	DOC_0005	Yes	Governance	Neutral	0.85

Modular LLM classification stages run across parallel partitions over the input rows (high mean confidence), producing the labelled dataframe.



Screenshots - Signal Results (Spreadsheet)

Excel signal_results.xlsx

D2 fx Yes – the company has a public multi-year net-zero portfolio target

	A	B	C	D	E	F
1	signal_id	signal_name	answer	evidence_summary	confidence	reviewed
2	SIG_01	Net-Zero Commitment	Yes	Public multi-year net-zero portfolio target.	0.94	✓
3	SIG_03	Circular Economy Focus	Yes	Portfolio holdings scaling closed-loop programmes.	0.88	✓
4	SIG_04	Supply-Chain Exposure	Yes	Elevated indirect-emissions exposure in some holdings.	0.81	✓
5	SIG_07	Regulatory Pressure	Yes	Emerging disclosure obligations apply to holdings.	0.86	✓
6	SIG_09	Green Financing Activity	No	No sustainability-linked financing identified.	0.72	✗
7	SIG_12	Emissions Disclosure Gap	Yes	Indirect-emissions methodology absent for several holdings.	0.78	✓
8	SIG_14	Renewable Investment	Yes	Recent capital allocated to clean-energy assets.	0.85	✓

signal_answers results summary

Each signal answer is grounded in supporting evidence with a confidence score and the analyst's review decision.



Screenshots - Pipeline Run (Terminal)

```
analyst@workstation — pipeline run

$ python -m pipeline.run --stage all --company "Client A"

[12:41:02] config    loaded pipeline configuration
[12:41:02] scope     signal set loaded for the active run
[12:41:03] precheck  wrote review workbook - 12 candidates
[12:41:03] review    ■ manual review gate - awaiting reviewed workbook
[12:48:20] review    ► resumed - 2 candidates excluded
[12:48:24] kb        ingested knowledge-base chunks ✓
[12:48:24] trigger   trigger evaluation complete 18.9s
[12:48:43] web       web evidence retrieval complete 196.4s
[12:52:00] chunk     hybrid retrieval + rerank complete 32.1s
[12:52:47] summary   signal & solution summaries written
[12:52:47] ✓ run complete - validated signals recorded · wall-clock=4m 21s

$
```

A complete strictly-sequential run with a mandatory human-review gate:
candidate signals narrow to a validated subset in $\approx 4\text{m } 21\text{s}$.



Conclusion

- A modular, retrieval-grounded, human-supervised GenAI engine reliably turns unstructured evidence into ranked, explainable recommendations.
- Met targets: $\approx 91\%$ classification F1, nDCG@10 of 0.81, a validated subset of signals, fully source-traceable outputs.

Future Scope

- Real-time, incremental (upsert-based) ingestion and continuous monitoring.
- Learned / feedback-calibrated scoring and larger gold-standard evaluation.
- Interactive dashboards and generalization beyond sustainability.



- ❶ Lewis et al., “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks,” *NeurIPS*, 2020.
- ❷ Brown et al., “Language Models are Few-Shot Learners,” *NeurIPS*, 2020.
- ❸ Robertson & Zaragoza, “The Probabilistic Relevance Framework: BM25 and Beyond,” *FnTIR*, 2009.
- ❹ Karpukhin et al., “Dense Passage Retrieval for Open-Domain QA,” *EMNLP*, 2020.
- ❺ Nogueira & Cho, “Passage Re-ranking with BERT,” *arXiv:1901.04085*, 2019.
- ❻ Cormack et al., “Reciprocal Rank Fusion,” *ACM SIGIR*, 2009.
- ❼ Manning et al., *Introduction to Information Retrieval*, Cambridge Univ. Press, 2008.



Thank You

